

# Retrieval over an Amended Law:

## Corpus Versioning, Paragraph-Level Citation and Judge Validation for Question Answering on the EU AI Act

Aghasalim Mustafazada

Howest University of Applied Sciences, Kortrijk, Belgium  
salim.mustafazada@student.howest.be

Murad Eyvazov

University of Massachusetts Boston, Boston, USA

October 2026

### Abstract

Question answering tools over legislation are usually built once and evaluated once. The law then changes under them. This paper measures what that does to retrieval, using Regulation (EU) 2024/1689 (the AI Act) and its amendment by Regulation (EU) 2026/1744. The corpus is versioned by the date of the text: the original Official Journal text of 12 July 2024 and the consolidated text of 27 July 2026. On 11 questions that only the amended text can answer, hybrid retrieval over the stale corpus returns the right article number for 63.6% of questions but the current wording for 0.0%. On the current corpus the figures are 81.8% and 63.6%. Article-level hit rate hides a stale corpus. Cutting the text into one chunk per numbered paragraph raises full recall on the 33 original answerable questions from 69.7% to 75.8% and mean reciprocal rank from 0.790 to 0.843, and 90.9% of those questions retrieve a gold paragraph. Citation precision is 0.176 against a ceiling of 0.232 at six retrieved passages. The question set grows to 123, with every gold reference and quote checked by a program against the legal text; the 67 new questions are easier than the original 45. The LLM judge used to grade answers agrees with 41 hand labels on 82.9% of answers (Cohen’s kappa 0.52), and accuracy is 66.7% strict and 69.7% lenient under either set of grades. Generation has not been re-run on the amended text. Code and results are released as an installable package.

**Keywords:** EU AI Act; Digital Omnibus; retrieval-augmented generation; legal information retrieval; consolidated legislation; evaluation; LLM-as-a-judge

**JEL classification:** K24, O33

## 1 Introduction

A retrieval-augmented generation (RAG) system answers a question from passages it first retrieves from a fixed collection of documents [8]. For legal questions this design is attractive, because the answer can cite the provision it rests on. A reader can check the citation. The weak point is the collection. When the law is amended, a system indexed on the old text keeps answering, and its citations still look correct.

In earlier work, one of us measured whether the claims in an answer are supported by the retrieved documents, across 31 languages [11]. That is groundedness. This paper asks the question one step earlier: are the retrieved documents still the law? An answer can be fully

grounded in a superseded text. A groundedness check will then pass it, and the reader will be misled.

The case is the EU Artificial Intelligence Act, Regulation (EU) 2024/1689 [3]. On 27 July 2026, Regulation (EU) 2026/1744, called the Digital Omnibus on AI, entered into force and amended it [4]. Among other changes it moved the high-risk dates, added prohibitions to Article 5, replaced Article 4 and added Articles 4a and 60a. The system studied here was built and measured on the 2024 text before the amendment. That gives a natural experiment: the same retriever, the same questions and two versions of the same law.

The paper is an evaluation study. It reports five things, all measured and all in a public repository [10]:

1. Corpus versioning by date of text, and a measurement of what a stale corpus does on 11 questions that only the amended text can answer. Article-level metrics look acceptable on the stale corpus. A check on the current wording shows it fails on every question.
2. Paragraph-level chunking and citation, compared with article-level chunking on the same questions.
3. A question set grown from 45 to 123, with every gold reference and every supporting quote checked by a program against the text.
4. A validation of the LLM judge, the citation parser and the abstention rule against 41 hand-labelled answers.
5. An installable package that reproduces every retrieval number without an API key.

The readership in mind is people who build or buy compliance tools, and people who study them. The method is simple on purpose. Most of the numbers need no language model to reproduce.

## 2 Related work

LegalBench [5] measures legal reasoning by large language models over many task types, but on fixed texts. Magesh et al. [9] evaluate commercial legal research tools and find that retrieval does not remove hallucination. Work on LLM judges [15] shows they agree with humans often but not always, which is why the judge here is checked against hand labels. None of these, to our knowledge, measures what happens to a legal RAG system when the law it indexes is amended. That is the gap this study addresses, at small scale.

## 3 Background: the AI Act and its 2026 amendment

Regulation (EU) 2024/1689 was published in the Official Journal on 12 July 2024 [3]. In the parsed version used here it has 113 articles, 180 recitals and 13 annexes. The recitals are the non-binding “whereas” paragraphs that explain the rules. The articles hold the rules.

Regulation (EU) 2026/1744 was adopted on 8 July 2026, published on 24 July 2026 and entered into force on 27 July 2026, the third day after publication under its Article 4 [4]. It amends 2024/1689 directly, in 43 numbered points of its Article 1. The Publications Office of the European Union publishes a consolidated text of 2024/1689 dated 27 July 2026, CELEX 02024R1689-20260727 [12]. That text has no legal effect. Only the Official Journal texts are authentic. For a question answering tool the consolidated text is still the right source, because it

is what a reader needs. Every change relied on in this paper was checked against the amending act itself.

The changes that matter for the questions below are these:

- **Dates (Article 113).** Chapter III, Sections 1 to 3, applies from 2 December 2027 for Article 6(2) and Annex III systems, and from 2 August 2028 for Article 6(1) and Annex I systems. The two new Article 5 points, (ba) and (bb), and Article 5(1a) and (1b) apply from 2 December 2026. The general date of 2 August 2026 did not change.
- **New prohibitions (Article 5).** Article 5(1)(ba) and (bb) add prohibitions concerning non-consensual intimate material of identifiable people and child sexual abuse material.
- **AI literacy (Article 4).** Replaced. It now concerns measures to support AI literacy and states that providers and deployers need not guarantee any specific level of AI literacy of any individual.
- **New articles.** Article 4a on processing special categories of personal data for bias detection and correction (Article 10(5) is deleted and its basis moved here), and Article 60a on real-world testing of products covered by Section B of Annex I.
- **Small mid-caps (SMCs).** New definitions in Article 3, simplified technical documentation under Article 11(1), and a fine cap under the new Article 99(6a) at whichever of the two amounts is lower.
- **Transitional rule (Article 111(4)).** Systems placed on the market before 2 August 2026 must comply with the Article 50(2) marking duty by 2 December 2026.

Most of these changes rewrite an existing article without renumbering it. That detail drives the main result. A system that checks only whether “Article 5” was retrieved cannot tell the 2024 Article 5 from the 2026 Article 5.

The consolidated text ships without a preamble, so the 180 recitals of the original are carried over unchanged. Of the 306 units of the original text, 248 parse to identical text in both versions.

## 4 System

The system is a standard hybrid RAG pipeline. It is described briefly here; the repository holds the full method [10].

**Source and parsing.** Both texts are fetched as XHTML from the Cellar API of the Publications Office, with CELEX 32024R1689 for the original and 02024R1689-20260727 for the consolidated text. The two use different markup. The parser reads both and rebuilds the nested lettered lists, which are encoded as tables, into an outline.

**Article-level chunks.** The first chunking follows the document’s own units: one chunk per article, recital or annex, split at numbered items and then sentences when a unit exceeds the 512-token limit of the encoder. Each chunk starts with a breadcrumb of chapter, section and article title. This gives 464 chunks for the 2024 text and 498 for the 2026 text (119 articles, 180 recitals and 14 annexes).

**Paragraph-level chunks.** The second chunking makes one chunk per numbered paragraph, or per top-level point where an article has no numbered paragraphs, as in Article 3 and Article 113. Each line carries its place in the outline, so a chunk knows that it is Article 5(1)

and holds points (f) to (h). The text is the same; only the cuts differ. This gives 921 chunks for 2024 and 994 for 2026.

**Retrieval.** Dense retrieval uses `BAAI/bge-small-en-v1.5` [14] in a Chroma store. Sparse retrieval is BM25 [13]. The two rankings are combined by reciprocal rank fusion with the usual constant of 60 [2]. Fusion uses ranks only, so no score scaling has to be fitted to the test set. Recitals are down-weighted by a factor of 0.5 in the fusion, because in earlier runs they crowded binding articles out of the top results. Ties are broken on chunk id, so the ranking is deterministic across machines. All results below use the hybrid retriever at  $k = 6$  unless stated.

**Generation and judging.** Answers come from `openai/gpt-oss-20b` through an OpenAI-compatible API. The prompt allows the model to answer `NOT_IN_CORPUS`. Answers are graded by a second model from a different family, `qwen/qwen3.6-27b`, which rates correctness against a reference answer as correct, partial or incorrect, and scores faithfulness claim by claim. A rule-based parser extracts bracketed citations, which feed a citation validity metric.

## 5 Evaluation design

### 5.1 Question sets

Table 1 lists the three question sets. All were written by the author.

**Table 1:** Question sets. “Original” was written against the 2024 text before the amendment. The other two were written from the 2026 consolidated text.

| set                                   | questions | single hop | multi hop | unanswerable | notes                                                            |
|---------------------------------------|-----------|------------|-----------|--------------|------------------------------------------------------------------|
| original ( <code>qa_set</code> )      | 45        | 21         | 12        | 12           | 33 answerable; m04 superseded, s15 and m12 incomplete after 2026 |
| amended ( <code>qa_amended</code> )   | 11        | 9          | 2         | 0            | answer only in the 2026 text                                     |
| extended ( <code>qa_extended</code> ) | 67        | 61         | 6         | 0            | 13 quote wording absent from the 2024 text                       |
| total                                 | 123       | 91         | 20        | 12           | 110 answerable for the 2026 text once m04 is removed             |

The **original** 45 questions include 12 that the Act cannot answer, such as questions about the GDPR or the Digital Services Act, so that refusal is measured. Several target places where nearly right is wrong, such as the three deadlines for serious incident reports in Article 73, or the switch from “whichever is higher” to “whichever is lower” for fines on SMEs in Article 99. The amendment makes three of them out of date. They are marked but left unchanged, because they are correct for the text the published numbers were measured on.

The **amended** 11 questions each rest on wording that exists only in the 2026 text: the two new high-risk dates, Article 5(1)(ba) and its start date, the new Article 4, Article 4a, Article 60a, Article 6(1a), Article 111(4), Article 99(6a) and Article 11(1) for SMCs. Each question records the exact quote its answer rests on. A test checks that every quote is inside a gold unit of the 2026 corpus and in no chunk of the 2024 corpus.

The **extended** 67 questions cover high-risk classification and obligations (28), prohibited practices (7), general-purpose AI models (7), transparency (5), penalties (6), dates and tran-

sitional periods (6), and provisions added in 2026 (8). Each has gold article and paragraph references, a short gold answer and the exact wording it rests on.

## 5.2 Programmatic checks against the text

A script, `eval/check_qa.py`, checks every set against the corpus and runs in the test suite. Every gold reference must exist in the corpus. Every quote must sit inside a gold paragraph. Every gold paragraph of the extended set must hold a quote. The script also reports which questions quote wording that the 2024 text lacks. On the current commit all three sets pass: 0 of the original 45, all 11 amended questions and 13 of the 67 extended questions quote such wording. This check does not prove a question is good. It proves the gold labels point at text that exists and says what the question claims.

## 5.3 Metrics

All retrieval metrics are computed at  $k = 6$  over the answerable questions.

- **Hit rate:** at least one gold article is in the top  $k$ .
- **Full recall:** every gold article is in the top  $k$ . This matters for multi-hop questions, where finding one of two articles produces an answer that reads as complete and is not.
- **MRR:** mean reciprocal rank of the first gold article.
- **Evidence hit:** every gold quote appears, ignoring case and whitespace, inside some top- $k$  chunk. This is the metric that tells a current provision from a superseded one with the same number.
- **Paragraph hit, paragraph full recall, paragraph MRR:** the same as above, with each retrieved chunk read as a citation of the paragraph it is labelled with. Article 5(1)(f) and 5(1)(h) count as the same paragraph; 6(3) and 6(4) do not. An article-level chunk can only be cited as the whole article.
- **Citation precision:** the share of distinct cited paragraphs in the top  $k$  that are gold.

nDCG [6] is also computed and is in the results files.

For answers, the judge reports strict accuracy (correct only), lenient accuracy (correct or partial), faithfulness, and correct and false abstention. Citation validity is the share of cited provisions that were in the retrieved set.

## 5.4 Judge validation

The answer grades come from an LLM judge, a practice that needs checking [15]. We graded 41 answers by hand from the text of the Act, without looking at the judge’s grade. They are every non-empty answer that did not abstain in two runs of the same generator, with answers identical across the runs counted once: 26 from the published run and 15 from an earlier run with a lower token cap. For each answer we recorded a grade and the provisions it cites as a reader sees them. Agreement is reported as raw agreement and Cohen’s kappa [1]. The abstention rule (the answer contains `NOT_IN_CORPUS`) is checked against our reading of all 45 rows of the published run. All of this runs on stored outputs and needs no API key.

## 6 Results

### 6.1 Baseline on the original text

On the 33 answerable original questions and the 2024 text, with article chunks, hybrid retrieval reaches a hit rate of 90.9% and full recall of 69.7%, against 81.8% and 51.5% for dense retrieval alone and 84.9% and 63.6% for BM25 alone. Multi-hop questions are the weak case: hybrid finds a gold article for all 12 but every gold article for only 41.7%.

Generation was measured once, on the 2024 text with article chunks (Table 2). The system refused all 12 unanswerable questions and its citations always pointed at retrieved passages. It also refused 21.2% of answerable questions.

**Table 2:** Answer quality on the original 45 questions, 2024 text, article chunks. Generator `openai/gpt-oss-20b`, judge `qwen/qwen3.6-27b`. Not re-run on the 2026 text.

| metric                                           | value         |
|--------------------------------------------------|---------------|
| answer accuracy, strict (33 answerable)          | 66.7%         |
| answer accuracy, lenient                         | 69.7%         |
| single-hop / multi-hop accuracy, strict          | 76.2% / 50.0% |
| faithfulness (claims entailed by retrieved text) | 90.2%         |
| citation validity                                | 100%          |
| correct abstention on unanswerable questions     | 100% (12/12)  |
| false abstention on answerable questions         | 21.2%         |

Of the 45 outcomes, 33 are classed as fine. Of the other 12, three are complete retrieval misses, three partial retrieval, four false abstentions, one a generation error and one an incomplete answer.

### 6.2 A stale corpus on amended questions

Table 3 is the central result. On the original questions the two texts give almost the same numbers, which is expected: the amendment leaves most of those provisions alone. On the 11 amended questions the picture splits.

**Table 3:** Hybrid retrieval, article chunks,  $k = 6$ , on both texts. “Current wording” is evidence hit. The MRR of 0.795 on the first row is the committed baseline; see Section 9 on the 0.790 figure.

| questions | text searched | $n$ | hit rate | full recall | MRR   | current wording |
|-----------|---------------|-----|----------|-------------|-------|-----------------|
| original  | 2024-07-12    | 33  | 90.9%    | 69.7%       | 0.795 | n/a             |
| original  | 2026-07-27    | 33  | 90.9%    | 69.7%       | 0.785 | n/a             |
| amended   | 2024-07-12    | 11  | 63.6%    | 45.5%       | 0.636 | 0.0%            |
| amended   | 2026-07-27    | 11  | 81.8%    | 63.6%       | 0.773 | 63.6%           |

Measured the usual way, by article number, the stale corpus answers 7 of 11 amended questions with a correct-looking provision. A hit rate of 63.6% would pass many sanity checks. Yet none of the retrieved chunks holds the current wording. An answer built on them would state the old rule with a citation that looks right. Table 4 shows this per question.

Three patterns stand out. First, a03, a05, a08, a09 and a10 are full hits on the old text by article number and misses by wording. These are the rewritten articles. Second, a06 and a07 ask about Articles 4a and 60a, which did not exist in 2024; the old corpus cannot return them at all. Third, a01 and a02, the two plain date questions, miss Article 113 on both texts. The top six results are all articles about high-risk systems, such as Articles 6, 8 and 43, and

**Table 4:** Per question, hybrid, article chunks,  $k = 6$ . A = gold article in top 6; F = all gold articles; W = current wording (evidence hit). P = gold paragraph in top 6 with paragraph chunks on the 2026 text.

| id  | gold paragraphs  | 2024 text |     |    | 2026 text |     |     | 2026 para. |
|-----|------------------|-----------|-----|----|-----------|-----|-----|------------|
|     |                  | A         | F   | W  | A         | F   | W   | P          |
| a01 | 113(c)(i)        | no        | no  | no | no        | no  | no  | no         |
| a02 | 113(c)(ii)       | no        | no  | no | no        | no  | no  | no         |
| a03 | 5(1)(ba)         | yes       | yes | no | yes       | yes | yes | yes        |
| a04 | 5(1)(ba), 113(a) | yes       | no  | no | yes       | no  | no  | yes        |
| a05 | 4(1)             | yes       | yes | no | yes       | yes | yes | yes        |
| a06 | 4a(2)            | no        | no  | no | yes       | yes | yes | yes        |
| a07 | 60a(1)           | no        | no  | no | yes       | yes | yes | yes        |
| a08 | 6(1a), 6(1b)     | yes       | yes | no | yes       | yes | yes | yes        |
| a09 | 111(4)           | yes       | yes | no | yes       | yes | yes | yes        |
| a10 | 99(6a)           | yes       | yes | no | yes       | yes | yes | yes        |
| a11 | 113(c)(i), 11(1) | yes       | no  | no | yes       | no  | no  | yes        |

not the article that holds the dates. This is a retrieval weakness that the amendment did not cause, but it hits the questions a compliance reader is most likely to ask.

### 6.3 Paragraph-level chunking and citation

Table 5 compares the two chunkings. On the original questions, paragraph chunks raise article-level full recall from 69.7% to 75.8%, which is two more questions, and MRR from 0.790 to 0.843. On 33 questions the difference is not statistically significant. It did, however, cost nothing, and it makes paragraph citation possible: 90.9% of questions now retrieve a gold paragraph. Article chunks score zero on paragraph metrics by construction, since an article chunk can only be cited as the whole article. The only article-level gold reference, s02 on the general date in the unnumbered text of Article 113, is missed by both chunkings.

**Table 5:** Article against paragraph chunks, hybrid,  $k = 6$ . Article-level columns: hit rate, full recall, MRR. Paragraph-level columns: hit, full recall, MRR, citation precision (CP).

| questions, text | chunks    | hit   | full rec. | MRR   | p. hit | p. full rec. | p. MRR | CP    |
|-----------------|-----------|-------|-----------|-------|--------|--------------|--------|-------|
| original, 2024  | article   | 90.9% | 69.7%     | 0.790 | 0.0%   | 0.0%         | 0.000  | 0.000 |
| original, 2024  | paragraph | 97.0% | 75.8%     | 0.843 | 90.9%  | 75.8%        | 0.757  | 0.176 |
| amended, 2026   | article   | 81.8% | 63.6%     | 0.773 | 0.0%   | 0.0%         | 0.000  | 0.000 |
| amended, 2026   | paragraph | 81.8% | 63.6%     | 0.818 | 81.8%  | 54.5%        | 0.773  | 0.155 |

Citation precision looks low, and it has a low ceiling. Most questions have one or two gold paragraphs, and the retriever returns six. A perfect ranker would score 0.232 on the original set and 0.212 on the amended set at  $k = 6$ . At  $k = 3$  the system scores 0.308 on the original set against a ceiling of 0.465. Citation precision should therefore be read against its ceiling, or the number of passages shown to the reader should be lowered.

On the amended questions, paragraph chunks leave article-level hit rate and full recall unchanged but lift paragraph hit rate to 81.8%. The two multi-hop questions, a04 and a11, are not fully recalled at article level, but each gets at least one gold paragraph.

## 6.4 The larger question set

Table 6 reports the extended set and all sets together on the 2026 text.

**Table 6:** Hybrid, 2026 text,  $k = 6$ . “All” is the three sets less m04, 110 answerable questions. Paragraph full recall and citation precision (CP) are paragraph level.

| questions | chunks    | $n$ | hit rate | full recall | MRR   | p. full recall | CP    |
|-----------|-----------|-----|----------|-------------|-------|----------------|-------|
| new 67    | article   | 67  | 95.5%    | 89.6%       | 0.812 | 0.0%           | 0.000 |
| new 67    | paragraph | 67  | 98.5%    | 92.5%       | 0.865 | 86.6%          | 0.171 |
| all       | article   | 110 | 92.7%    | 80.9%       | 0.799 | 0.9%           | 0.002 |
| all       | paragraph | 110 | 96.4%    | 83.6%       | 0.853 | 79.1%          | 0.169 |

The new questions are clearly easier than the original 45. The likely reason is that they were written with the paragraph open, so they reuse its words, which is what BM25 rewards. The original set was written to be hard and still is. The extended set is best read as a regression check for the paragraph chunker and the amended text, not as an estimate of performance on real users’ questions. Paragraph chunks beat article chunks on all of it, by 2.7 points of full recall on the 110. Of the 13 extended questions that need the amended wording, 10 retrieve it in the top six with paragraph chunks and 11 with article chunks.

## 7 Judge validation

Table 7 compares the judge and the two rule-based components with the hand labels.

**Table 7:** Automatic components against 41 hand-labelled answers (45 rows for the abstention rule).

| component                                  | items | agreement    | Cohen’s $\kappa$ |
|--------------------------------------------|-------|--------------|------------------|
| judge grade, correct / partial / incorrect | 41    | 82.9%        | 0.52             |
| judge grade, correct against not correct   | 41    | 82.9%        | 0.48             |
| abstention rule (NOT_IN_CORPUS)            | 45    | 100%         | 1.00             |
| citation parser, exact match per answer    | 41    | 80.5%        |                  |
| citation parser, precision / recall        | 41    | 100% / 79.6% |                  |

A kappa of 0.52 is moderate agreement on the usual scale [7]. The disagreements have a clear shape. All 7 are about the partial grade. The judge rated 5 answers partial that we rated correct: four for leaving out a detail of the reference that the question did not need (s13, s21, m01, m10), and one for a sentence cut off after the answer was complete (s17). It rated m04 correct twice where we rated it partial: the list of articles is right, but the answer describes Articles 102 to 109 and 112 wrongly, and the judge only compares against the reference. The judge is strict about omissions and blind to wrong extra claims. Every answer that either rater called incorrect, both did (5 cases).

On the published run the disagreements cancel. Replacing the judge’s grades with ours where we have them, strict accuracy stays at 66.7% and lenient accuracy at 69.7%. The headline accuracy numbers are therefore not an artefact of the judge, though the agreement on single answers is only moderate.

The citation parser never invents a citation, but it misses one in five. Most misses are citations outside brackets, such as an article number in bold or in running text. Two are brackets left unclosed by an answer that was cut off. The 100% citation validity in Table 2 is therefore computed over bracketed citations only.

Two things are not validated. The claim-level faithfulness judge has not been checked against hand labels, which would need each claim read against the retrieved passages. The judge has also not been re-run live on the 41 answers, because no API key was available on the machine used for this study. The repository includes a command that does both comparisons once a key is set.

## 8 Discussion: stale corpora and compliance tools

The main finding is about measurement, not about this system. A legal RAG tool evaluated by whether the right article number comes back will look healthy on a superseded corpus. On the amended questions here, the stale corpus scores 63.6% on that metric and 0% on the wording. The reason is structural. Amending acts mostly rewrite provisions in place. Article 5 stays Article 5 after two prohibitions are added to it, and Article 113 stays Article 113 after its dates change. A citation-based metric, and a reader who checks citations, will both be satisfied.

For compliance use this is the worst kind of error. Consider a provider asking when the high-risk obligations for an Annex III system start. A tool indexed on the 2024 text has no passage that contains “2 December 2027”. If it answers, it answers from the old Article 113, with a correct citation. An earlier groundedness check [11] would pass that answer, since it is supported by the retrieved text. The legal research tools studied by Magesh et al. [9] show that citation and correctness come apart even in commercial products; a stale corpus is one simple way for that to happen.

Three practical points follow from the results.

1. **Version the corpus by the date of the text and show the date.** Here each chunk carries a version field, there is one index per version, and the application defaults to the latest text with a selector for the older one. A reader should see which version an answer comes from.
2. **Test on wording, not only on numbers.** The evidence hit metric is cheap: it needs a quote per question and a string match. It caught a failure that hit rate, recall and MRR all missed. Questions that only the new text can answer, with their quotes checked against both versions, make a small but sharp regression test for each amendment.
3. **Cite at the level a lawyer cites.** Paragraph chunks gave paragraph citations at no cost to article-level retrieval. They also make it possible to say which paragraph changed, which an article-level tool cannot.

The date questions deserve a separate note. Both a01 and a02 fail on both texts. The question names high-risk systems, so the retriever returns the articles about high-risk systems; the dates are in a final provision with no topical words. Transitional and final provisions are exactly where amendments tend to land, and exactly what a compliance reader needs. A tool that retrieves well on substance and poorly on application dates will be most wrong when the law has just changed.

Finally, on the 33 original answerable questions the 2024 corpus gives the same hit rate and full recall as the 2026 corpus. A team that re-ran only its existing test set after the amendment would see no change and conclude the corpus was fine. Keeping a test set current with the law is part of keeping the tool current.

## 9 Limitations

- **Generation was not re-run on the amended text.** No API key was available on the machine used for this study. All answer-level numbers (Table 2) were measured on the 2024 text with article chunks. The effect of a stale corpus is shown at retrieval level only. There is no answer-level comparison yet.
- **Small sets, one author.** The amended set has 11 questions and the original set 33 answerable ones. Small differences, such as the two-question gain in full recall from paragraph chunks, are not statistically significant. All questions and all hand labels were written by the author, who also built the system. No second annotator was available, so inter-annotator agreement among humans is not measured.
- **The extended set is easy.** Its questions reuse the wording of the paragraphs they were written from, which favours BM25. Its numbers flatter retrieval.
- **Judge coverage.** The faithfulness judge was not validated, and the live re-judge was not run. The citation parser misses about one citation in five, so citation validity covers bracketed citations only.
- **Consolidated text.** The 2026 corpus is a consolidated text with no legal effect. The changes used here were checked against the amending act, but the corpus itself is not authentic law. The recitals are carried over from 2024.
- **No recital questions.** No question has a recital as its gold answer, which flatters the recital down-weighting.
- **English only.** The Act is equally authentic in 24 languages. One was tested.
- **A small MRR discrepancy.** The article-chunk baseline on the original questions and the 2024 text has an MRR of 0.795 in the committed baseline file and 0.790 in the retrieval-only and article-against-paragraph runs, with every other article-level metric equal. The repository records that near-ties in fusion ordering differed across machines before a tie-break fix. Table 5 uses 0.790 so that both rows come from the same kind of run.
- **Not legal advice.** The system is a research artefact. It should not be used to make compliance decisions.

## 10 Reproducibility

Code, question sets, hand labels, per-question results and figures are at <https://github.com/aghasalim/eu-ai-act-rag>. The numbers in this paper come from commit 24d507fe57d1d1575f670259bc4096516a1b4a08 (9 October 2026), which contains the merged pull requests #7 to #13. The repository lists a Zenodo DOI, 10.5281/zenodo.23003624 [10].

The code installs as the Python package `euactrag` 0.1.0 (Python 3.12), with a command line tool and a two-function API. It is not on PyPI; the wheel and source archive are built by `make dist` and by continuous integration on every push. Retrieval results need no API key:

- `make setup && make corpus && make index`: fetch, parse and index both texts.
- `make eval-retrieval`: the original baseline.
- `make eval-versions`: Tables 3 and 4.

- `make eval-paragraph`: Table 5.
- `make eval-extended`: Table 6.
- `make check-qa`: the programmatic checks of all 123 questions.
- `make agreement`: Table 7, from stored outputs.

Answer generation (`make eval`) needs an API key for an OpenAI-compatible endpoint. Every figure in the repository README is recomputed from the per-question records by independent implementations in several languages, and continuous integration fails if they disagree.

The corpus is reproduced from the EU Publications Office under the Commission’s reuse policy. The project is not affiliated with or endorsed by the European Union.

## References

- [1] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37-46, 1960.
- [2] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009. pp. 758-759.
- [3] European Parliament and Council of the European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, OJ L, 2024/1689, 12.7.2024. CELEX 32024R1689, 2024. URL <http://data.europa.eu/eli/reg/2024/1689/oj>.
- [4] European Parliament and Council of the European Union. Regulation (EU) 2026/1744 amending Regulation (EU) 2024/1689 (Digital Omnibus on AI). Official Journal of the European Union, OJ L, 2026/1744, published 24.7.2026, in force 27.7.2026, 2026.
- [5] Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Ré, et al. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. arXiv:2308.11462.
- [6] Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422-446, 2002.
- [7] J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159-174, 1977.
- [8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020. pp. 9459-9474.
- [9] Varun Magesh, Faiz Surani, Matthew Dahl, Mirac Suzgun, Christopher D. Manning, and Daniel E. Ho. Hallucination-free? Assessing the reliability of leading AI legal research tools. arXiv:2405.20362, 2024.

- [10] Aghasalim Mustafazada. EU AI Act RAG: source code, question sets and results (euactrag 0.1.0). GitHub repository, commit 24d507f, 2026. URL <https://github.com/aghasalim/eu-ai-act-rag>.
- [11] Aghasalim Mustafazada and Teymur Eyvazov. A multilingual groundedness detector that runs on a Raspberry Pi: Trained token classification against LLM judges and English-only detectors in 31 languages. SSRN, 2026. URL <https://doi.org/10.2139/ssrn.7497398>.
- [12] Publications Office of the European Union. Regulation (EU) 2024/1689, consolidated text of 27 July 2026. CELEX 02024R1689-20260727. Retrieved from the Cellar API. Consolidated texts have no legal effect, 2026.
- [13] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333-389, 2009.
- [14] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, et al. C-Pack: Packaged resources to advance general Chinese embedding. arXiv:2309.07597, 2023.
- [15] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.